

Signal-Driven AI Systems



EMG-Based Muscle Health Profiling & Athletic Assessment

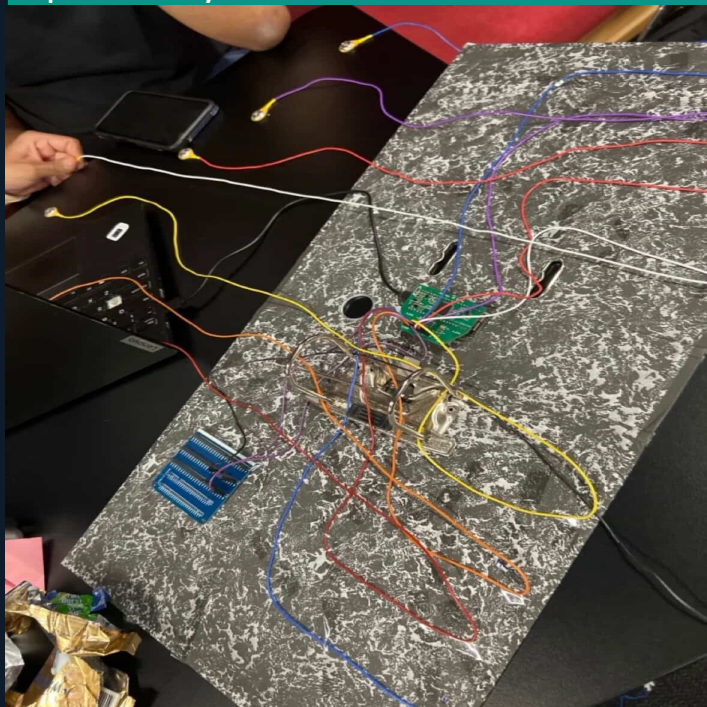
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Hardware & Interface

OpenBCI Cyton Hardware



OpenBCI GUI Demo



The Problem

Athletes and sports players have no affordable, accessible tool to objectively assess their own muscle health.



**Expensive access to
top-tier hardware**



**No comparison
baseline**



Insights stay raw

What We're Building

An EMG-based muscle health profiling & athletic suitability assessment system

Participant attends once · electrodes applied · forearm grip protocol · AI-generated personalised report



Muscle Strength Score

Percentile for their age group



Fatigue Resistance

vs. population norms



Athletic Classification

Sedentary → Elite profiling



Sport Suitability

Match to competitive level

Research Question

Can ML models accurately assess and classify muscle health, strength, and fatigue resistance from a single EMG session

and how do these profiles differ across age groups, sports participation levels, and lifestyle factors?

Study design

Type	Cross-sectional - one session per participant
Participants	30-50 people across 4 age groups
Who	Sports players, gym-goers, sedentary individuals
Duration	20-30 minutes per session

Project Objectives

- 01** Data Collection of >30 participants across 4 defined age groups
- 02** Develop a robust signal cleaning pipeline with <5% artefact retention.
- 03** Train classifier for achieving >75% accuracy.
- 04** Deliver an analytical AI chatbot

Related Work & Literature Review

What existing research tells us about EMG-based fatigue detection:

EMG fatigue detection is scientifically well-established

Phinyomark et al. (2012) and De Luca (1997) demonstrated that median frequency (MDF) of the EMG power spectrum reliably decreases as a muscle fatigues - the gold-standard non-invasive fatigue marker. RMS amplitude also changes predictably with fatigue onset, giving two complementary indicators to train on.

Machine learning can classify EMG fatigue states accurately

Oskoei & Hu (2007) showed that time- and frequency-domain features extracted from windowed EMG signals - including RMS, ZCR, and MDF - are sufficient to train classifiers (Random Forest, SVM) with accuracy above 80% on controlled datasets. This directly validates our feature extraction approach.

Age-group differences in EMG fatigue patterns are documented

Narici et al. (2004) showed that older adults exhibit faster MDF decline rates and lower peak RMS values compared to younger individuals under equivalent contractions - directly justifying our cross-age-group comparison as a scientifically grounded research question.

Conclusion: The science is proven. What is missing is accessibility - no tool brings EMG + ML + plain-English AI together for regular athletes.

State of the Art

Academic EMG Research

Phinyomark et al., De Luca

- Strong feature extraction methods
- Well-documented in literature
- Lab-only, not accessible
- No AI interpretation layer

Commercial EMG Tools

Delsys, Noraxon, BioMed

- Clinical-grade accuracy
- Well-validated hardware
- \$5,000–\$50,000+ cost
- No plain-English output

Consumer Fitness AI

Whoop, Apple Watch, Garmin

- Accessible and affordable
- Good UX
- No EMG - surface data only
- No muscle-specific analysis

Our Novelty

No existing tool combines all three of these in one accessible system:



Accessible EMG hardware

OpenBCI Cyton - affordable, portable, research-grade



ML classification

Automated fatigue + athletic profile scoring

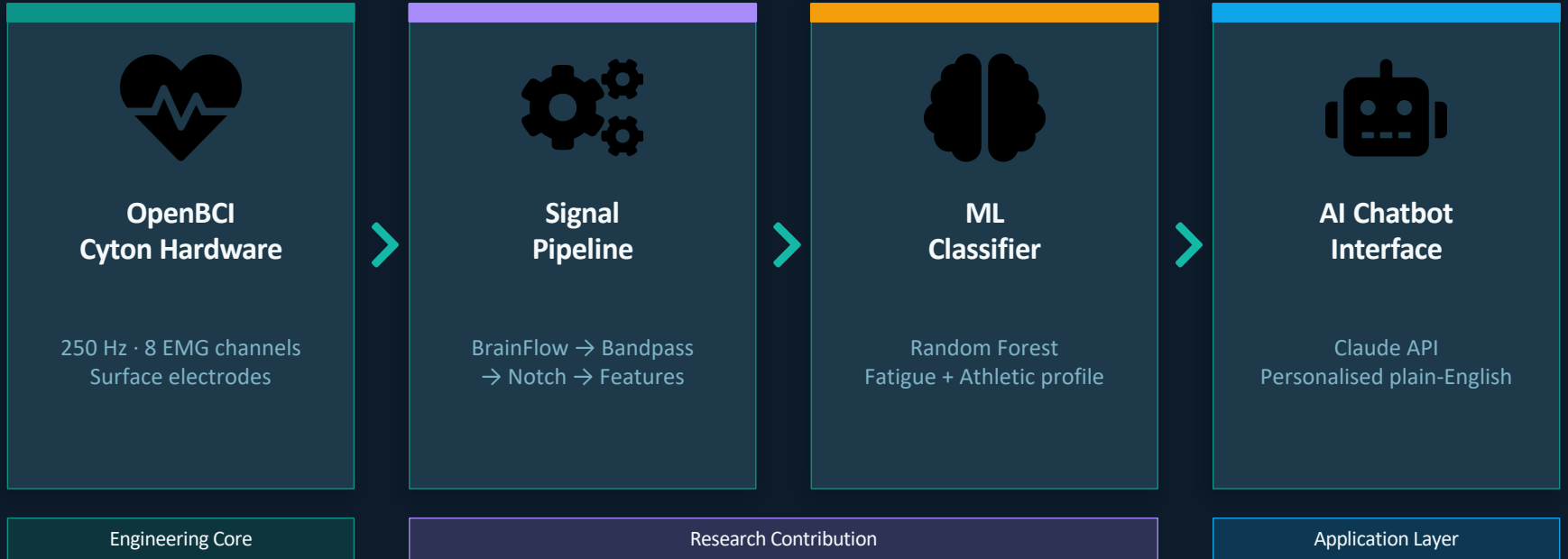


Conversational AI interface

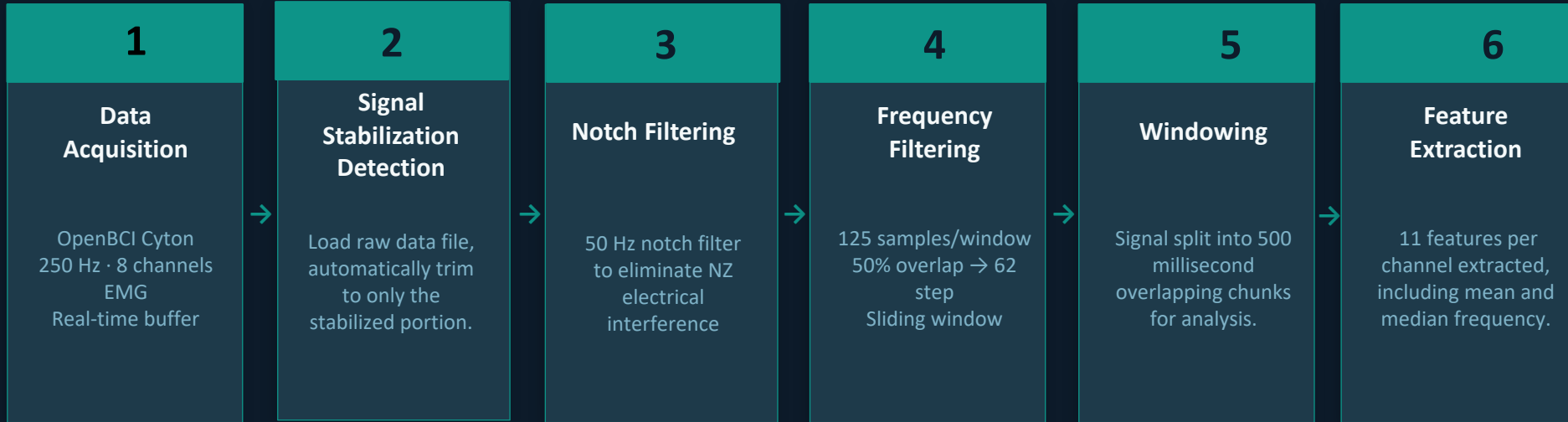
Plain-English results via AI API

This combination does not exist in any current academic, clinical, or consumer product.

System Architecture



Signal Processing Pipeline

**Key insight:**

MDF (median frequency) is the primary fatigue indicator - it shifts downward as muscles tire. The rate of MDF decline (MDF_slope) across windows is the strongest predictor in the classifier.

Exercise Protocol

Sustained Isometric Grip Contraction

- 1 Participant seated, arm resting on table, elbow at 90°
- 2 Two surface EMG electrodes on forearm flexor (underside, 1/3 down from elbow)
- 3 Ground electrode on elbow bone (bony landmark)
- 4 Squeeze forearm grip trainer at ~70% max effort - hold continuously
- 5 Continue until failure or 60 seconds, then rest 30 seconds
- 6 Repeat for 3 sets - Set 1 = Fresh · Set 2 = Moderate · Set 3 = Fatigued

ML Labelling

- | | |
|-------|------------|
| Set 1 | → Fresh |
| Set 2 | → Moderate |
| Set 3 | → Fatigued |

Why this works

- Reliably induces fatigue in 30–60 sec
- Forearm flexors produce strong EMG signal
- Grip trainer standardizes resistance across all participants
- Safe, repeatable, no gym equipment required
- MDF shifts visibly as muscle fatigues

Machine Learning - Classification Targets

1**Fatigue State**

3-class

Fresh

Moderate

Fatigued

Labelled by time-window position in session

2**Athletic Profile**

4-class

Non-Active

Recreational

Trained

Elite

Labelled from participant intake form

3**Sport Suitability**

4-class per sport

Below avg

Average

Above avg

Elite

Scored per sport category and age group

Scoring & Algorithm Selection

Results are expressed as percentiles within age group - not raw numbers.

< 25th
Below average

25–50th
Average

50–75th
Above average

75–90th
Strong

90th+
Elite

Example output:

"Your fatigue resistance is in the top 25% for your age group, which is consistent with competitive-level rugby players."

Algorithms

★ Random Forest

Primary - handles mixed features, outputs importance scores

SVM

Comparison - strong for high-dimensional feature spaces

K-Nearest Neighbours

Comparison - intuitive baseline, easy to interpret

Research Question

Can machine learning models accurately detect and classify muscle fatigue using features extracted from EMG biosignals, and do fatigue patterns vary across different age groups?

Methodology

- Extract RMS, MDF, ZCR, MAV, WL per 500 ms window of cleaned signal
- Label windows: Set 1=Fresh, Set 2=Moderate, Set 3=Fatigued
- Train Random Forest - compare against SVM and KNN
- Group by age band (18-25, 26-40, 41-60, 61+) and compare MDF decline rates
- Statistical tests to confirm age-group differences

Expected Outcomes

- >75% accuracy on 3-class fatigue classification
- Feature importance: which EMG signals best predict fatigue
- Evidence of age-group differences in fatigue onset patterns
- Comparison table: Random Forest vs SVM vs KNN
- Validated model ready for chatbot integration

Research Question

How can signal processing techniques effectively clean and preprocess multi-channel EMG signals from the OpenBCI Cyton Board to enable reliable muscle fatigue detection?

Methodology

- Record raw EMG from forearm flexor at 250 Hz via OpenBCI Cyton
- Apply 50 Hz notch filter (iirnotch) - removes NZ powerline noise
- Apply 4th-order Butterworth bandpass 20-120 Hz
- Slice into 500 ms windows with 50% overlap
- Prove effectiveness: SNR and MDF stability before vs after filtering

Expected Outcomes

- Cleaned, validated EMG dataset for ML classifier
- Quantified SNR improvement with before vs after plots
- Documented, repeatable preprocessing pipeline
- preprocess.py integrated into full system
- Signal quality report confirming <5% artefact retention

Research Question

How can AI-driven analysis of EMG biosignals be used to develop a real-time system that monitors muscle activity and provides actionable insights for athletes?

Methodology

- UI/UX fully designed in Figma - complete (session history, profile, hardware status)
- Implement frontend in HTML/CSS/JS connected to live ML backend
- Integrate Claude API - inject actual ML scores as context for every response
- Results delivered as percentiles in plain English, not raw signal values
- User testing with real participants to confirm insights are actionable

Expected Outcomes

- Functional chatbot answering questions about the user's specific EMG session
- AI reads actual ML output - not generic EMG knowledge
- User testing confirms athletes understand and act on results
- Real-time: results available immediately after session ends
- Complete UI with session history, profile, and hardware status

Scrum Development Methodology

Scrum Master

Aryan Shah

Sprint planning, stand-ups, blocker removal, meeting facilitation

Product Owner

Rayyan Abzal

Backlog management, acceptance criteria, supervisor liaison

Developer

Maalav Garg

Signal pipeline, ML model development, unit testing

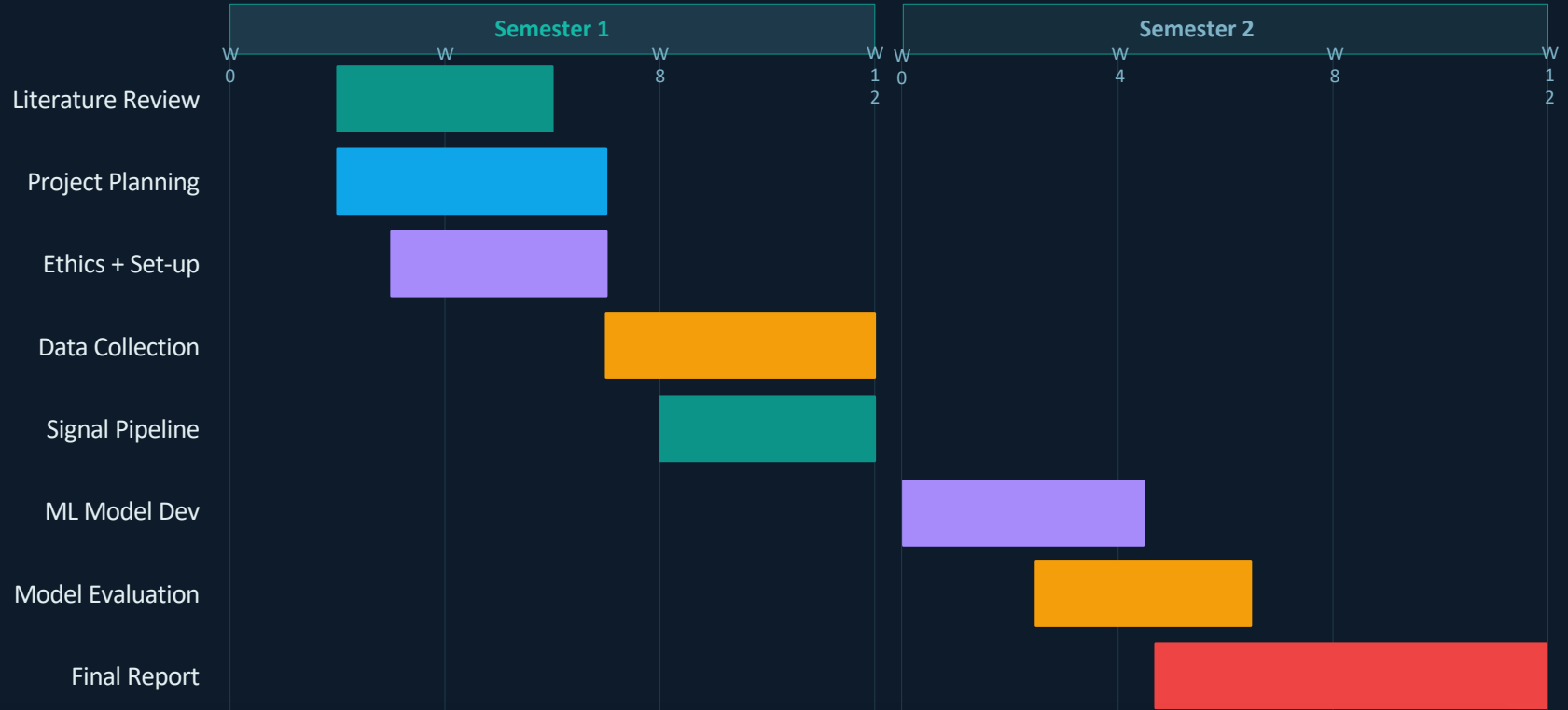
Sprint Structure

- 2-week sprint cycles once we start development
- Sprint review with supervisor each fortnight
- Backlog: pipeline · ML · frontend · data collection
- Definition of done: tested, documented, merged

Key Agile Practices

- Weekly stand-ups (in-person or Teams)
- Trello Project Management
- Git version control
- Iterative delivery - working software each sprint

Project Timeline



Ethics approval is the critical path item - all human data collection blocked until confirmed (target S1 Week 5–8)

Feasibility & Risk Assessment

Ethics & regulatory:

Human participant data collection requires AUT ethics confirmation. No data collected until approved. Informed consent, anonymisation, and secure storage will be applied. Confirming with supervisor.

High**Ethics approval delay**

Mitigation: Hardware issue being resolved with supervisor - pilot testing on group members once stable signal confirmed

Medium**Insufficient participants**

Mitigation: Target sports clubs, AUT gym, campus outreach from Week 7 onwards

Medium**Electrode noise / signal quality**

Mitigation: Standardised placement protocol, impedance check before every session

Medium**Class imbalance in dataset**

Mitigation: class_weight='balanced' in sklearn, SMOTE oversampling if needed

Summary



Cross-sectional EMG muscle health profiling - one session, real actionable insight for sports athletes



Random Forest classifier scoring participants against age-group and sport-specific population norms



AI chatbot delivering personalised plain-English assessments - powered by Claude API reading live ML output



Novel combination of OpenBCI hardware + classical ML + conversational AI - no equivalent tool currently exists